

# Null-Coalescing Assignment Operator in C# 8

In this article, I am going to discuss **Null-Coalescing Assignment Operator in C# 8** with Examples. Please read our previous article where we discussed [Indices and Ranges in C# 8](#) with Examples. C# 8.0 has introduced a new operator that is known as a Null-Coalescing Assignment Operator (??=).

## Null-Coalescing Assignment Operator (??=) in C# 8

C# 8.0 introduces the null-coalescing assignment operator **??=**. We can use this **??=** operator to assign the value of its right-hand operand to its left-hand operand only if the left-hand operand evaluates to null. That means the null-coalescing assignment operator **??=** assigns a variable only if it is null. The syntax is given below.

**a ??= b;**

Here, a is the left and b is the right operand of the null-coalescing operator **??=**. If the value of a is null, then the **??=** operator assigns the value of b in a. If the value of a is not-null, then it does not evaluate b.

It simplifies a common coding pattern where a variable is assigned with a value if it is null. For a better understanding, please have a look at the below diagram. Here, you can observe before C# 8, how we are checking null and assigning a value if it is null and how we can achieve the same in C# 8 using the null-coalescing assignment (**??=**) operator.

```
// C# 1..7
if (variable == null)
{
    variable = expression;
}

// C# 8
variable ??= expression;
```

## Points to Remember While working with ??= in C#:

1. The left-hand operand of the **??=** operator must be a variable, a property, or an indexer element.
2. It is right-associative.
3. You cannot overload **??=** operator.
4. You are allowed to use the **??=** operator with reference types and value types.

## Null-Coalescing Assignment Operator Example in C#:

Let us see an example for a better understanding. The following example is self-explained, so please go through the comment lines for a better understanding.

```

using System;
using System.Collections.Generic;
using System.Threading.Tasks;
namespace Csharp8Features
{
    class NullCoalescingAssignment
    {
        static void Main(string[] args)
        {
            //Creating null variables
            List<int>? numbersList = null;
            int? number = null;
            //The numbersList is null, so will create new List<int>() object
            numbersList ??= new List<int>();
            //As number is null, so it will assign 25 to number and store it in the list
            numbersList.Add(number ??= 25);
            //Now number is not null, so it will not assign 30 to number,
            //so the previous value 25 will store again in the list
            numbersList.Add(number ??= 30);
            foreach (var item in numbersList)
            {
                Console.Write(item + " "); // output:25 25
            }
            Console.WriteLine();
            Console.WriteLine(number); // output:25
        }
    }
}

```

Output:

```

25 25
25

```

## Real-Time Use Cases of Null-Coalescing Assignment Operator

With the help of Null-Coalescing Assignment `??=` Operator, we can remove many redundant if-else statements and make our code more readable and understandable. Let us understand this with an example. Here, first, I will show you an example using the if statement, and then I will convert the same example using Null-Coalescing Assignment `??=` Operator so that you will get a better idea.

### Example using If Statements:

```

using System;
using System.Threading.Tasks;
namespace Csharp8Features
{
    class NullCoalescingAssignment
    {

```

```

static async Task Main(string[] args)
{
    // C# 1..7
    int? Age = null;
    if (Age == null)
    {
        Age = 20;
    }
    Console.WriteLine(Age);
}
}
}

```

**Same Example using Null-Coalescing Assignment ??= Operator:**

```

using System;
using System.Threading.Tasks;
namespace Csharp8Features
{
    class NullCoalescingAssignment
    {
        static async Task Main(string[] args)
        {
            // C# 8
            int? Age = null;
            Age ??= 20;
            Console.WriteLine(Age);
        }
    }
}

```

In the next article, I am going to discuss [Unmanaged Constructed Types in C# 8](#) with Examples. Here, in this article, I try to explain **Null-Coalescing Assignment in C# 8** with Examples. I hope you enjoy this Null-Coalescing Assignment in C# with Examples article.